

Recipe Retrieval pipeline

From the raw CSV to a ranked answer. Every model is used as downloaded and runs locally.

IS603
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1
INGEST

Parse and clean the Shengtao/recipe CSV

FIELD	WHAT WE DO	EXAMPLE
url	drop duplicate rows	32,722 rows to 32,719
prep, cook, total time	text to minutes	"1 hr 10 mins" to 70
ingredients	one row per line	kept as written
fractions	rewrite as plain text	"1 ½ cups" to "1 1/2 cups"
nutrition	keep 19 of 32 columns	13 dropped, over 99% empty
category, author	move to lookup tables	22 categories
directions	split into sentences	used to cut Step chunks

Models used while ingesting

MODEL	USED FOR
nomiic-embed-text	one vector per Chunk
	one vector per sentence
bert-base-uncased	token counts for fixed cuts

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DATABASE DESIGN

category

author

recipe

recipe_nutrition

recipe_ingredient

chunking_strategy

chunk

Source tables in 3NF
chunk is built from them
and never edited by hand

one partition per strategy

CHOICE

REASON

12 filter columns on chunk

one partition per strategy

Loaded_at flag

index filters only its own table

keeps BM25 counts apart

half-built ones stay hidden

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CHUNKING

One Recipe becomes

Summary
title, description

Ingredients
whole list

Step 1

Step 2

Step 3

same in every strategy

only this part changes

Every Chunk starts with the Recipe title. Chunks do not overlap.

STRATEGY	STEP	CHUNK CUT	STEPS	MEDIAN
semantic	topic change,	80 to 400 chars	101,000	185 ch
fixed	every 56 tokens		100,536	224 ch
sentence	every 3 sentences		110,314	178 ch

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CHOOSING MODELS

JOB	MODEL	SIZE	RUNS IN	INPUT
Embed text	nomiic-embed-text	768 dims	Ollama	prefix search_document: or search_query:
Rerank Hybrid results	mxbai-rerank-base-v1	184M params	sentence-transformers	query and Chunk together
Write test queries	qwen2.5:7b	7B params	Ollama	evaluation only

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INDEX

One ParadeDB index per partition holds

BM25 TEXT	content, with English stemming
EXACT	category, kind, strategy
NUMBERS	minutes, rating, calories, protein and 5 more
VECTOR	embedding, cosine, cluster index with probe 0.02

Vector index tested

INDEX	DENSE R@5	HYBRID R@5
ParadeDB cluster index USED	0.859	0.919
pgvector HNSW	0.872	0.916

One index serves BM25, vectors and filters together.

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SEARCH

query

embed
23 ms

BM25 top 50 18 ms

cosine top 50 32 ms

merge with RRF
k is 60, under 1 ms

rerank top 50
mxbai, 837 ms

top 5 Recipes

embed

BM25

cosine

rerank takes 91% of 921 ms

Sparse stops after BM25, Dense after cosine

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TESTING THE RERANKER

RERANKER	RECALL@5	MRR	RELATIVE TIME	SETTING	TRIED	KEPT
MiniLM-L6-v2	0.923	0.799	0.26	Reranker	4 models	mxbai
bge-reranker-base	0.919	0.758	1.00	Candidates reranked	20, 50, 100	50
mxbai-rerank-base-v1	0.966	0.895	1.25	RRF k	10, 60, 100	60
bge-reranker-v2-m3	0.943	0.838	2.90	Changed per test	one setting, same queries	

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EVALUATE

300 Recipes
seed 603

qwen2.5 writes queries
3 to 8 words

298 kept
57 fixed by hand

pooled answer key
see table below

score 4 configs
Recall@k, MRR, nDCG

ANSWER KEY

WHAT HAPPENS

COUNT

Pool

Judge

Score

Check

collect the top 5 Recipes of every config, strategy and test for each query

an LLM marks a pair right if the searcher would be as happy with it as with the source Recipe

a query counts at the rank of its first right Recipe, not only the one it came from

judge 120 pairs again, blind

5,428 pairs

691 right

180 queries gained one

110 agreed

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RESULTS

CONFIG	RECALL@5	MRR	MEDIAN TIME	QUESTION	ANSWER
Sparse, BM25	0.822	0.664	19 ms	Best quality	Hybrid with mxbai
Dense, cosine	0.859	0.697	69 ms	Fastest	Sparse
Fusion, RRF only	0.879	0.725	46 ms	Sparse wins on	exact words
Hybrid, bge-reranker-base	0.919	0.758	805 ms	Dense wins on	synonyms, like stew for soup
Hybrid, mxbai	0.966	0.895	1,010 ms	Chunking strategy	no clear winner

Rows 7 and 9 use the semantic strategy. Report run 20260928-122309, 298 queries. Row 9 times are medians of three timing runs on an M-series MacBook. The mxbai time is estimated from its test run, where it took 1.25 times as long as bge. Row 6 times come from that run.